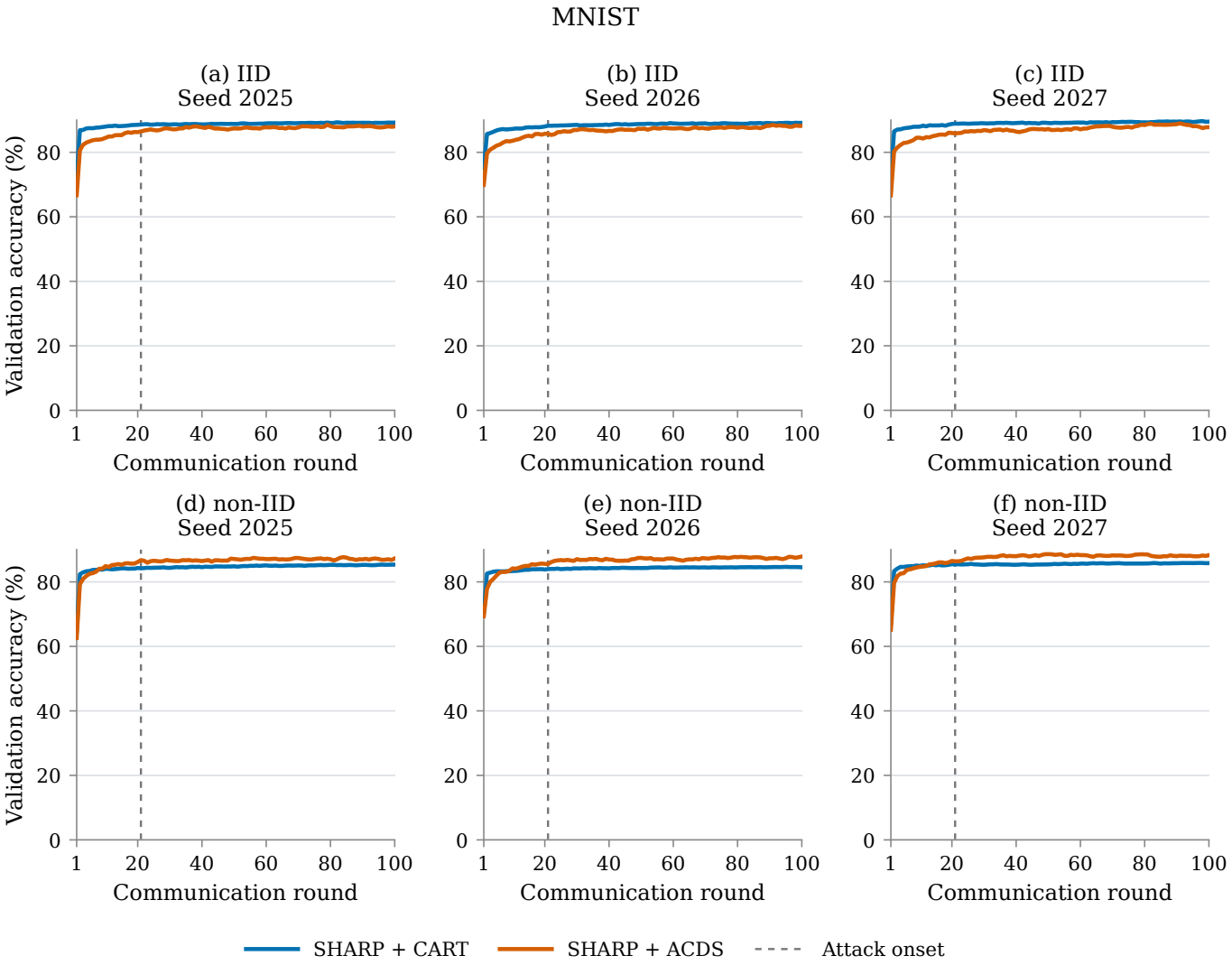
MNIST
(a) IID
Seed 2025
(b) IID
Seed 2026
(c) IID
Seed 2027
(d) non-IID
Seed 2025
(e) non-IID
Seed 2026
(f) non-IID
Seed 2027
Validation accuracy (%)
0
20
40
60
80
1
20
40
60
80
100
Communication round
SHARP + CART
SHARP + ACDS
Attack onset